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My Class Notes

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1. Avogadro's number
2. Ideal gas equation.
3. Deviation from ideal behaviour
4. liquefaction of gases
5. Critical temperature
6. Kinetic energy and molecular speeds (elementary idea)
7. Liquid State- vapour pressure
8. Viscosity and surface tension (qualitative idea only, no mathematical derivations)



1. Avogadro's number

A **mole** is a collection that contains the same number of particles as there are carbon atoms in 12.01 g of carbon.

6.022×10^{23} atoms of an element (**Avogadro's number**).

<u>1 mol of Element</u>		<u>Number of Atoms</u>
1 mol C	=	6.022×10^{23} C atoms
1 mol Na	=	6.022×10^{23} Na atoms
1 mol Au	=	6.022×10^{23} Au atoms



A mole

- of a *covalent compound* has Avogadro's number of molecules
$$1 \text{ mol CO}_2 = 6.022 \times 10^{23} \text{ CO}_2 \text{ molecules}$$
$$1 \text{ mol H}_2\text{O} = 6.022 \times 10^{23} \text{ H}_2\text{O molecules}$$
- of an *ionic compound* contains Avogadro's number of formula units
$$1 \text{ mol NaCl} = 6.022 \times 10^{23} \text{ NaCl formula units}$$
$$1 \text{ mol K}_2\text{SO}_4 = 6.022 \times 10^{23} \text{ K}_2\text{SO}_4 \text{ formula units}$$

Substance	Number and Type of Particles
1 mol of Al	6.022×10^{23} atoms of Al
1 mol of Fe	6.022×10^{23} atoms of Fe
1 mol of H ₂ O	6.022×10^{23} molecules of H ₂ O
1 mol of vitamin C (C ₆ H ₈ O ₆)	6.022×10^{23} molecules of vitamin C
1 mol of NaCl	6.022×10^{23} formula units of NaCl



Avogadro's number converts *moles* of a substance to the number of *particles*.

How many Cu atoms are in 0.50 mol of Cu?

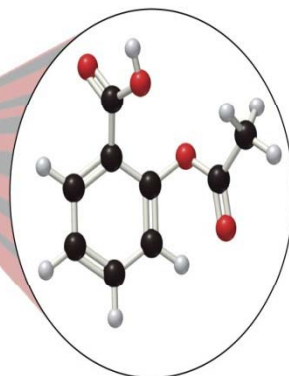
$$\begin{aligned} 0.50 \text{ mol Cu} \times \frac{6.022 \times 10^{23} \text{ Cu atoms}}{1 \text{ mol Cu}} \\ = 3.0 \times 10^{23} \text{ Cu atoms} \end{aligned}$$

Qr. How many moles of CO_2 are in 2.50×10^{24} molecules of CO_2 ?

Ans: 4.15 mol of CO_2



How many O atoms are in 0.150 mol of aspirin, $C_9H_8O_4$?



Aspirin $C_9H_8O_4$

3.61×10^{23} O atoms

Number of atoms in 1 molecule

Carbon (C)



Hydrogen (H)



Oxygen (O)





Characteristics of a Gas

- A) Gases assume the shape and volume of a container.
- B) Gases are the most compressible of all the states of matter.
- C) Gases will mix evenly and completely when confined to the same container.
- D) Gases have lower densities than liquids or solids.

Measurable Properties of a Gas

- A) Mass - moles - symbol is n
- B) Pressure - symbol is P
- C) Volume - symbol is V
- D) Temperature - must be in K - symbol is T

Units of pressure and their relationships are: $1 \text{ atm} = 760 \text{ mm Hg} = 760 \text{ torr} = 101.3$

$\text{kPa} = 14.7 \text{ psi} = 29.92 \text{ inches of Hg}$



- 1) The volume of a gas is directly proportional to the number of molecules (moles) of gas at the same T and P.**
- 2) One mole of any gas occupies the same volume for a given T and P and contains the same number of particles - atoms - molecules.**



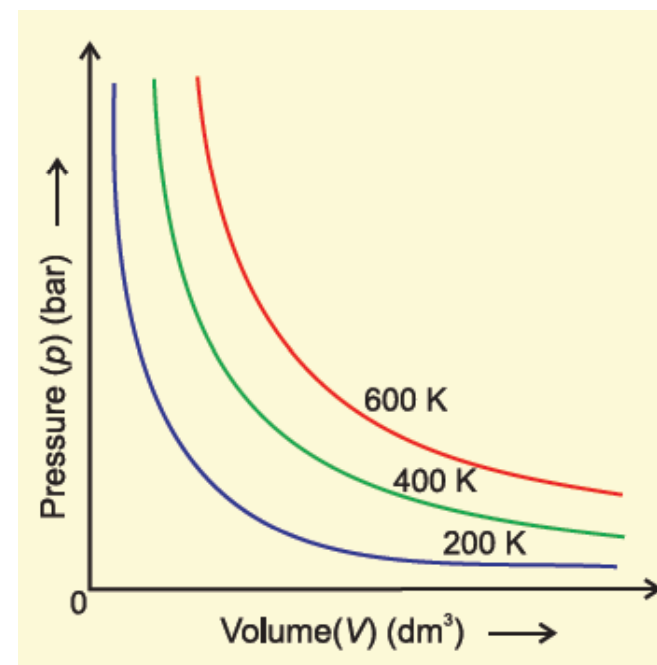
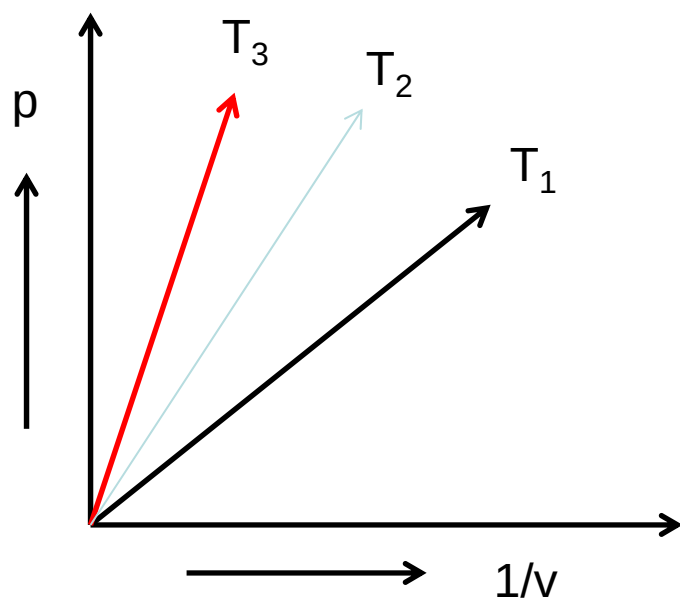
Boyle's Law: at constant temperature, the pressure of a fixed amount (i.e., number of moles n) of gas varies inversely with its volume. This is known as Boyle's law.

For a fixed amount of gas and constant temperature, $PV = \text{constant}$

$$p \propto \frac{1}{V} \quad (\text{at constant } T \text{ and } n) \quad p_1 V_1 = p_2 V_2 = \text{constant}$$

$$p = k_1 \frac{1}{V}$$

$$\frac{p_1}{p_2} = \frac{V_2}{V_1}$$



Each curve corresponds to different constant temperature called **ISOTHERM**



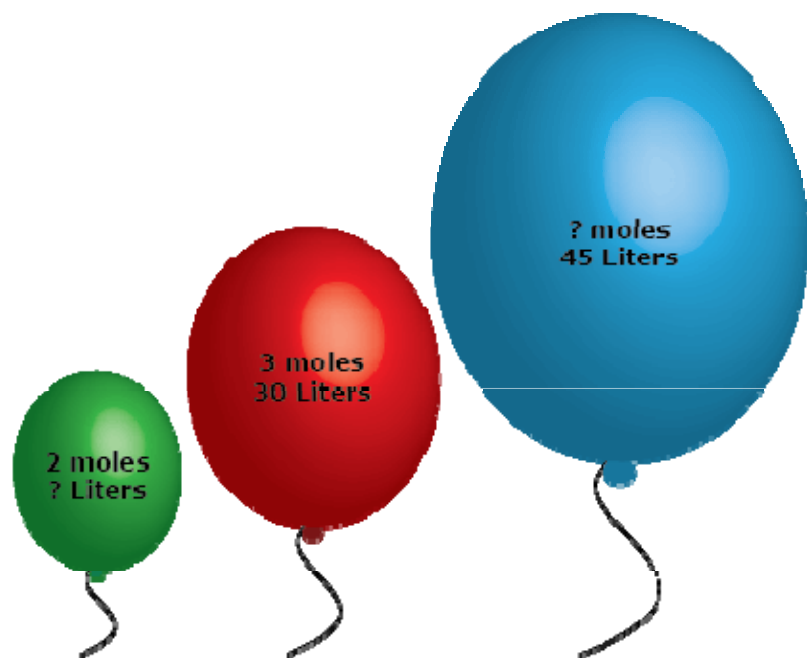
Charles's Law: at constant pressure the volume is linearly proportional to temperature.

$$V/T = \text{constant}$$



Avogadro's Law - EQUAL VOLUMES OF DIFFERENT GASES CONTAIN EQUAL NUMBERS OF MOLECULES WHEN MEASURED AT THE SAME TEMPERATURE AND PRESSURE.

For a fixed pressure and temperature, the volume of a gas is directly proportional to the number of moles of that gas. $V/n = k = \text{constant}$.





Gay-Lussac's Law

Lussac studied the direct relationship between temperature and pressure of a gas. As the temperature increases the pressure a gas exerts on its container increases. During his experiments volume of the system and amount of gas were held constant.



Dalton's Law

It states that the total pressure exerted by the mixture of non-reactive gases is equal to the sum of the partial pressures of individual gases

The sum of the partial pressures of the gases in a mixture = the total pressure
or $P = P_A + P_B + P_C + \dots$ where P_i = the partial pressure of component i.



2. Ideal gas equation

The three laws (1. Boyle's Law 2. Charles' Law 3. Avogadro's Law) which we have learnt till now can be combined together in a single equation which is known as **ideal gas equation**

Ideal gas law the functional relationship between the pressure, volume, temperature and moles of a gas.

At constant T and n ; $V \propto \frac{1}{p}$ **Boyle's Law**

At constant p and n ; $V \propto T$ **Charles' Law**

At constant p and T ; $V \propto n$ **Avogadro Law**

$PV = nRT$; all gases are ideal at low pressure.

$PV = nRT$. Each of the individual laws is contained in this equation.



Ideal Gases

An “ideal” gas exhibits certain theoretical properties. Specifically, an ideal gas ...

- Obeys all of the gas laws under all conditions.
- Does not condense into a liquid when cooled.
- Shows perfectly straight lines when its V and T & P and T relationships are plotted on a graph.

In reality, there are no gases that fit this definition perfectly. We assume that gases are ideal to simplify our calculations.



KINETIC MOLECULAR THEORY OF GASES <http://selfstudy.in/HSEPhysics/KineticTheoryOfGas.pdf>

Kinetic Theory Of Gas

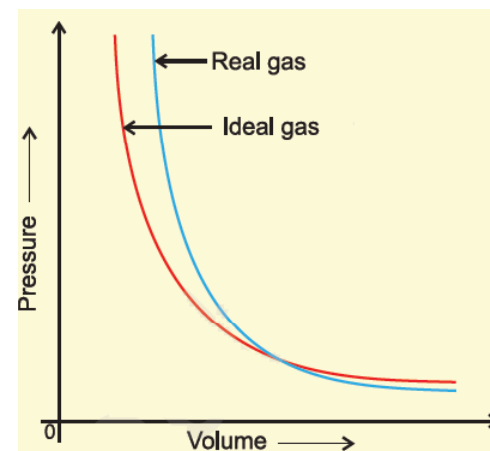
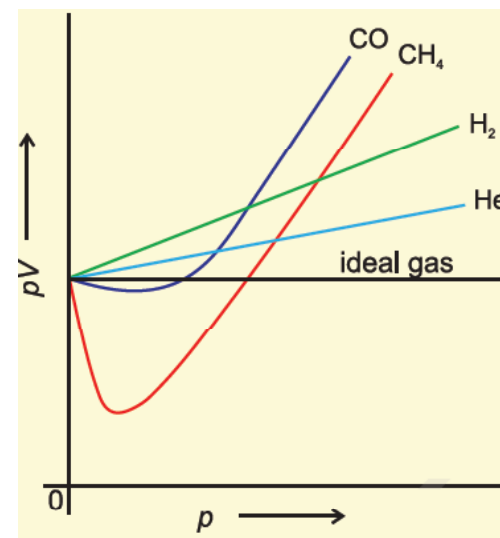


Postulates of Kinetic Theory: The molecules of a gas are assumed to obey the following properties.

Axioms (assumptions):

- (1) The molecules of a gas are perfectly elastic, hard, solid sphere identical in all respect such as mass, size form etc. The molecules of different gases are different.
- (2) The molecules of a gas are in a state of incessant random motion, moving in all directions with all possible velocities. In course of their motion the molecules collide with one another and with the wall of the containing vessel. But in the steady state the collisions do not affect the molecular density of the gas.
- (3) There is no intermolecular force of attraction between the molecules of the gas hence the gas molecules do not possess any potential energy. All the energy possessed by the gas molecules is kinetic energy.
- (4) The time for which two molecules remain in contact with each other is known as time of collision. The time of collision is negligible compared to the time during which the molecules are moving freely.
- (5) The distance covered by the molecule between any two collisions is known as free path.
- (6) All the free paths traversed by a molecule are not same, some are long and some are short. The average of all the free paths traversed by a molecule is known as mean free path λ .
- (7) The mean free paths of all the molecules of a gas are same. The gas whose molecules obey the postulates is said to be a perfect gas or an ideal gas. Under normal temperature and pressure no gas obeys these postulates and hence all gases are known as real gas. However under extremely low pressure and very high temperature molecules of all gases obey the above postulates and behave as ideal or perfect gas.

3. BEHAVIOUR OF REAL GASES: DEVIATION FROM IDEAL GAS BEHAVIOUR





We find that two assumptions of the kinetic theory do not hold good. These are

- (a) There is no force of attraction between the molecules of a gas.
- (b) Volume of the molecules of a gas is negligibly small in comparison to the space occupied by the gas.

If assumption (a) is correct, the gas will never liquify. However, we know that gases do liquefy when cooled and compressed.

Real gases show deviations from ideal gas law because molecules interact with each other.

Pressure Correction:

$$p_{ideal} = p_{real} + \frac{an^2}{V^2}$$

observedcorrection
pressureterm

Volume Correction: Gas molecules occupy volume hence gas instead of moving in volume V , it will have $V-nb$ volume to move.

Therefore we write

$$(P + an^2/v^2)(V-nb) = nRT$$

Above equation is known as **van der Waals equation.**



The Ideal Gas Law

$$PV = nRT$$

P = Pressure (in kPa)

T = Temperature (in K)

V = Volume (in L)

n = moles

$$R = \frac{8.31 \text{ kPa} \cdot \text{L}}{\text{K} \cdot \text{mol}}$$

R is constant. If we are given three of P, V, n, or T, we can solve for the unknown value.

From Boyle's Law:

$$P_i V_i = P_f V_f \quad \text{or } PV = \text{constant}$$

From combined gas law:

$$P_i V_i / T_i = P_f V_f / T_f \quad \text{or } PV/T = \text{constant}$$



Developing the ideal gas law equation

$PV/T = \text{constant}$. What is the constant?

At STP: $T = 273\text{K}$, $P = 101.3\text{ kPa}$, $V = 22.4\text{ L/mol}$

Because V depends on mol, we can change equation to:

Mol is represented by n , constant by R :

Rearranging, we get:

$$PV = nRT$$

$$\text{At STP: } (101.3\text{ kPa})(22.4\text{ L}) = (1\text{ mol})(R)(273\text{K})$$

$$R = \frac{8.31\text{ kPa} \cdot \text{L}}{\text{K} \cdot \text{mol}}$$

$$PV = \text{constant}$$

$$\frac{PV}{T \cdot \text{mol}}$$

$$PV = R$$

$$\frac{PV}{Tn}$$

Note: always use kPa, L, K, and mol in ideal gas law questions (so units cancel)



4. LIQUIFACTION OF GASES: Real gases behave in the same manner as carbon dioxide

5. Critical temperature :

At 73 atmospheric pressure, liquid carbon dioxide appears for the first time. The temperature 30.98°C is called **critical temperature (T_c) of carbon dioxide**.

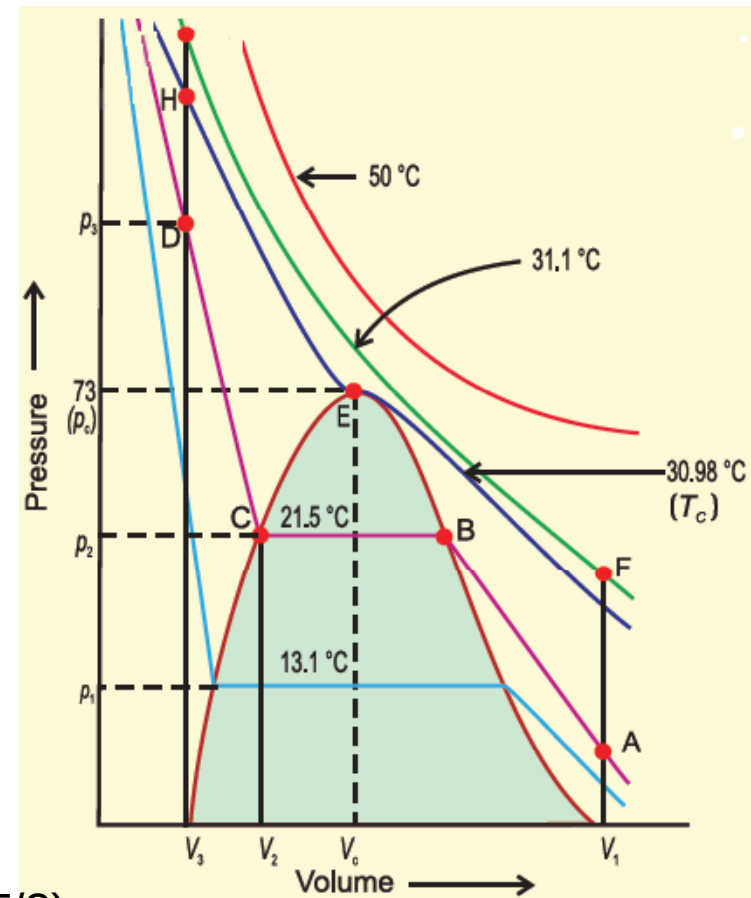
This is the highest temperature at which liquid carbon dioxide is observed. Above this temperature it is gas. Volume of one mole of the gas at critical temperature is called **critical volume (V_c) and pressure** at this temperature is called **critical pressure (p_c)**.

The critical temperature, pressure and volume are called critical constants.

6. Kinetic energy and molecular speeds

Refer the link for KE of Gas Molecules ($3KT/2$)

<http://selfstudy.in/HSEPhysics/PressureExertedByGas.pdf>





7. Liquid State- vapour pressure:

If an evacuated container is partially filled with a liquid, a portion of liquid evaporates to fill the remaining volume of the container with vapour. Initially the liquid evaporates and pressure exerted by vapours on the walls of the container (vapour pressure) increases. After some time it becomes constant, an equilibrium is established between liquid phase and vapour phase. Vapour pressure at this stage is known as **equilibrium vapour pressure or saturated vapour pressure**.

Since process of vapourisation is temperature dependent; the temperature must be mentioned while reporting the vapour pressure of a liquid.

Temperature at which vapour pressure of liquid is equal to the external pressure is called boiling temperature at that pressure. Vapour pressure of some common liquids at various temperatures is given in. At 1 atm pressure boiling temperature is called **normal boiling point**.

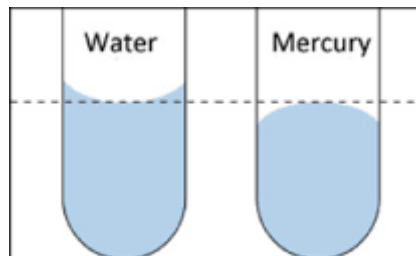
If pressure is 1 bar then the boiling point is called **standard boiling point of the liquid**. Standard boiling point of the liquid is slightly lower than the normal boiling point because 1 bar pressure is slightly less than 1 atm pressure. The normal boiling point of water is $100\text{ }^{\circ}\text{C}$ (373 K), its standard boiling point is $99.6\text{ }^{\circ}\text{C}$ (372.6 K)



8. Viscosity and surface tension (qualitative idea only, no mathematical derivations)

Surface Tension:

A molecule in the bulk of liquid experiences equal intermolecular forces from all sides. The molecule, therefore does not experience any net force. But for the molecule on the surface of liquid, net attractive force is towards the interior of the liquid, due to the molecules below it. Since there are no molecules above it



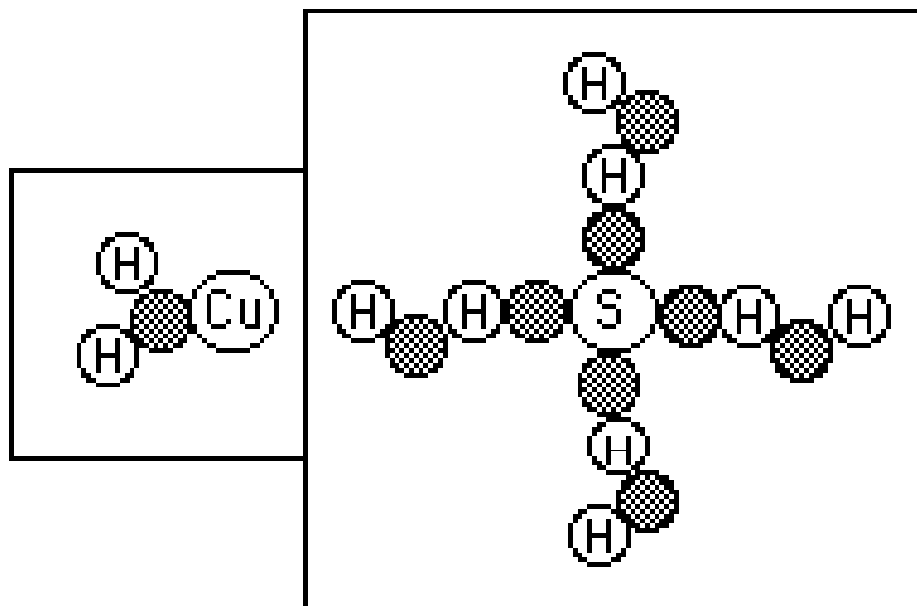
Refer Link For Viscosity: <http://selfstudy.in/HSEPhysics/NewtonsLawViscosity.pdf>

Details about Viscosity and Surface Tension we will study in Physics.



Hydrates

- Sometimes there are gaps within the ions in the lattice structure, leaving room for water molecules to join. The ratio of water molecules to the other ions depends on the size of the ions, the charge of the ions, and the type of lattice
- Such ionic compounds that can bind to water molecules are called hydrates.





Diffusion the process whereby a gas spreads out through another gas to occupy the space with uniform partial pressure.



Effusion the process in which a gas flows through a small hole in a container.